Drug Database Full Screening — Hardware Configuration Spec

Compute Requirements Analysis

Task	Bottleneck	Scale
12 Database Downloads	Network I/O	~155 GB compressed
SQLite/TSV parsing	CPU single core	ChEMBL ~30GB SQLite
Molecular fingerprint computation	CPU multi-core	~10^7 molecules × 2048-bit Morgan
Yajie multi-expert inference	GPU	~10^4 drugs × ~10^4 targets = 10^8 pairs
Results storage	Disk I/O	~10 GB final results

Plan A: Minimum Viable (Using Existing Equipment)

Component	Configuration	Already Have?
CPU	Any i7/Ryzen 7	
GPU	1× RTX 4090 24GB	
Memory	32GB	
Storage	External 4TB HDD	¥800
Total Budget		**¥800**

Bottleneck: Only 1 GPU. 10^8 pairs serial inference takes several days.

Plan B: Production Configuration (Recommended)

Component	Model	Price	Purpose
CPU	AMD Ryzen 9 9950X (16C/32T)	¥4,500	Molecular fingerprint parallel computation

Component	Model	Price	Purpose
GPU	2× RTX 4090 24GB	¥26,000	Yajie multi-expert parallel inference
Memory	64GB DDR5-6000	¥2,000	Large SQLite memory mapping
Motherboard	X670E (dual PCIe 5.0 x8)	¥2,500	Dual GPU support
System Disk	1TB NVMe PCIe 4.0	¥500	OS + Python
Data Disk	2× 4TB NVMe PCIe 4.0 (RAID0)	¥6,000	Database storage + high-speed read/write
PSU	1600W 80+ Platinum	¥2,000	Dual 4090 full load ~900W
Cooling	360mm AIO liquid cooling	¥800	9950X full load ~200W
Case	Full-tower E-ATX	¥800	Dual GPU space
Total		**¥45,100**	

Performance Estimate: - Molecular fingerprints: 16 cores parallel, ~10ˆ7 molecules 2 hours - Yajie inference: 2× 4090 parallel, 10ˆ8 pairs 8-12 hours - Total end-to-end: **full run within 1 day**

Plan C: Full-Speed Configuration

Component	Model	Price	Purpose
CPU	AMD Threadripper 7960X (24C/48T)	¥12,000	Extreme parallel fingerprints
GPU	4× RTX 4090 24GB	¥52,000	4-way parallel Yajie
Memory	128GB DDR5 ECC	¥4,000	All databases in-memory ma
Motherboard	TRX50 (4× PCIe 5.0 x16)	¥6,000	4 GPU
System Disk	2TB NVMe PCIe 5.0	¥1,500	
Data Disk	4× 4TB NVMe (RAID10)	¥12,000	Redundancy + high speed

Component	Model	Price	Purpose
PSU	Dual 1600W redundant	¥6,000	
Cooling	Custom water cooling	¥5,000	
Case	Server tower	¥2,000	
Total		**¥100,500**	

Performance Estimate: $4\times$ 4090 + 24 cores, Full Screening $<$ 4 hours.

Three-Plan Comparison

	A (Existing)	B (Recommended)	C (Full-Speed)
Budget	¥800	¥45,000	¥100,000
GPU	$1\times$ 4090	$2\times$ 4090	$4\times$ 4090
Full Screening Time	3-5 days	8-12 hours	$<$ 4 hours
Concurrent CC	1 way	3 ways	6+ ways
NEP Training			

What Not to Buy

Item	Rationale
A100/H100 80GB	3090/4090 24GB suffices. Yajie uses M small models, doesn't need large VRAM
Xeon	Single-socket Threadripper: more cores for less money
Professional GPU (RTX A6000)	4090 same chip, half price
NAS	Local NVMe RAID is faster
Cloud GPU (AWS/AutoDL)	¥5,000+ per run, better to buy cards